CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Faculty of Technology & Engineering

Devang Patel Institute of Advance Technology and Research (DEPSTAR)

CSUE301 – Big Data Analytics

Practical File – ODD Semester 2026-27

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Practical 1

Aim: To learn basic Linux commands for system information, file and directory management, file searching, permissions, backups, and system monitoring in a Linux environment.

Problem definition:

You have joined **Data Analytics Inc.** as a Linux System Administrator. Your task is to set up the **Marketing Analytics** project environment by creating directories, managing files, monitoring system resources, searching data, and creating backups for the Marketing Analytics team.

Objectives:

- Identify the current user and working directory.
- Display Linux system information.
- Create and manage directories and files.
- Add and view file contents.
- Search files and text using Linux commands.
- Copy, rename, move, and delete files.
- Create backups and monitor system resources.
- Understand basic Linux file management operations.

Observation / Output

1. Identify the current user.

```
yashu@LAPTOP-J824BISG:~$ whoami
yashu
```

2. Find the current working directory.

```
yashu@LAPTOP-J824BISG:~$ pwd
/home/yashu
```

3. Display system information.

```
yashu@LAPTOP-J824BISG: $ uname -a
Linux LAPTOP-J824BISG 6.6.87.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun  5 18:30:46 UTC 2025 x86_64 x86_64 x86_64 GNU
/Linux
```

4. Check system uptime and date.

```
yashu@LAPTOP-J824BISG:~$ uptime
 16:38:45 up 4 min,  1 user,  load average: 0.00, 0.04, 0.01
yashu@LAPTOP-J824BISG:~$ date
Wed Aug  5 16:38:48 UTC 2026
```

5. Create a directory named MarketingAnalytics

```
yashu@LAPTOP-J824BISG:~$ cd ~
yashu@LAPTOP-J824BISG:~$ mkdir MarketingAnalytics
yashu@LAPTOP-J824BISG:~$ ls
MarketingAnalytics  OS  development
```

6. Create subdirectories: RawData, ProcessedData, Reports.

```
yashu@LAPTOP-J824BISG: $ cd MarketingAnalytics
yashu@LAPTOP-J824BISG: /MarketingAnalytics$ ls
yashu@LAPTOP-J824BISG: /MarketingAnalytics$ mkdir RawData ProcessedData Reports
yashu@LAPTOP-J824BISG: /MarketingAnalytics$ ls
ProcessedData  RawData  Reports
yashu@LAPTOP-J824BISG: /MarketingAnalytics$
```

7. Create data and report files.

```
yashu@LAPTOP-J824BISG: /MarketingAnalytics/RawData$ touch report1.txt
yashu@LAPTOP-J824BISG: /MarketingAnalytics/RawData$ touch customer_data.txt
yashu@LAPTOP-J824BISG: /MarketingAnalytics/RawData$ cd ../Reports
yashu@LAPTOP-J824BISG: /MarketingAnalytics/Reports$ touch report1.txt
```

8. Add sample content.

```
yashu@LAPTOP-J824BISG: /MarketingAnalytics/RawData$ touch report1.txt
yashu@LAPTOP-J824BISG: /MarketingAnalytics/RawData$ touch customer_data.txt
yashu@LAPTOP-J824BISG: /MarketingAnalytics/RawData$ cd ../Reports
yashu@LAPTOP-J824BISG: /MarketingAnalytics/Reports$ touch report1.txt
yashu@LAPTOP-J824BISG: /MarketingAnalytics/Reports$ cd ../RawData
yashu@LAPTOP-J824BISG: /MarketingAnalytics/RawData$ nano customer_data.txt
yashu@LAPTOP-J824BISG: /MarketingAnalytics/RawData$ cd ../Reports
yashu@LAPTOP-J824BISG: /MarketingAnalytics/Reports$ nano report1.txt
yashu@LAPTOP-J824BISG: /MarketingAnalytics/Reports$ cat report1.txt
Marketing Analytics Report
```

9. Display file contents.

```
yashu@LAPTOP-J824BISG: /MarketingAnalytics/RawData$ cat report1.txt
Marketing Analytics Report

Sales increased by 18%

Prepared by Yashesh Amin
yashu@LAPTOP-J824BISG: /MarketingAnalytics/RawData$ cat ../RawData/customer_data.txt
CustomerID,Name,City
101,Raj Ahmedabad
102,Priya,Vadodara
103,Amit,Surat
```

10. List all files and directories.

```
yashu@LAPTOP-J824BISG:~/MarketingAnalytics/Reports$ ls -l
total 8
-rw-r--r-- 1 yashu yashu 73 Aug  5 16:50 customer_data.txt
-rw-r--r-- 1 yashu yashu 77 Aug  5 16:54 report1.txt
yashu@LAPTOP-J824BISG:~/MarketingAnalytics/Reports$ ls -R
.:
customer_data.txt  report1.txt
```

11. Search for report files.

```
yashu@LAPTOP-J824BISG:~/MarketingAnalytics$ find . -name "*.txt"
./Reports/report1.txt
./Reports/customer_data.txt
./RawData/report1.txt
./RawData/customer_data.txt
yashu@LAPTOP-J824BISG:~/MarketingAnalytics$ find . -name "report*"
./Reports/report1.txt
```

12. Search specific content inside files.

```
yashu@LAPTOP-J824BISG:~/MarketingAnalytics$ grep "Sales" Reports/report1.txt
Sales increased by 18%
yashu@LAPTOP-J824BISG:~/MarketingAnalytics$ grep "Raj" RawData/customer_data.txt
101,Raj Ahmedabad
```

13. Copy report files to Backup directory.

```
yashu@LAPTOP-J824BISG:~/MarketingAnalytics$ mkdir Backup
yashu@LAPTOP-J824BISG:~/MarketingAnalytics$ cp Reports/report1.txt Reports/Marketing_Report.txt
yashu@LAPTOP-J824BISG:~/MarketingAnalytics$ ls Reports
Marketing_Report.txt  customer_data.txt  report1.txt
```

14. Rename files.

```
yashu@LAPTOP-J824BISG:~/MarketingAnalytics$ mv Reports/report1.txt Reports/Markerting_Report.txt
yashu@LAPTOP-J824BISG:~/MarketingAnalytics$ ls Reports
Markerting_Report.txt  Marketing_Report.txt  customer_data.txt
```

-

15. Move files between directories.

```
yashu@LAPTOP-J824BISG:~/MarketingAnalytics$ mv RawData/customer_data.txt ProcessData
yashu@LAPTOP-J824BISG:~/MarketingAnalytics$ ls ProcessData
ProcessData
```

16. Delete a file.

```
yashu@LAPTOP-J824BISG:~/MarketingAnalytics$ rm RawData/report1.txt
yashu@LAPTOP-J824BISG:~/MarketingAnalytics$ ls RawData
```

17. Delete an empty directory.

```
yashu@LAPTOP-J824BISG:~/MarketingAnalytics$ rmdir Backup
yashu@LAPTOP-J824BISG:~/MarketingAnalytics$ ls
```

```
yashu@LAPTOP-J824BISG:~/MarketingAnalytics$ ls
ProcessedData  RawData  Reports
```

Result :

The Linux project environment was successfully created. Various Linux commands were executed to manage files and directories, search data, perform backup operations, and monitor system resources successfully.

Key Questions /Analysis:

Q1. What is the difference between absolute and relative paths?

- Absolute Path: Starts from the root directory (/) and gives the complete location of a file or directory.
- Relative Path: Starts from the current working directory and depends on the current location.

Q2. Why are Linux permissions important for file security?

Linux permissions protect files from unauthorized access, control read, write, and execute permissions, and improve system security and data integrity.

Q3. What is the purpose of the grep command?

The grep command searches for specific words or patterns inside one or more files and filters command output.

Q4. How does Linux support efficient file management?

Linux provides a hierarchical directory structure, powerful file management commands (ls, cp, mv, rm, find, grep), file permissions, and shell scripting for automation.

Q5. Why are backups important in enterprise environments?

Backups prevent permanent data loss, help recover files after failures or accidental deletion, protect against malware, and support business continuity.

Supplementary Problems:

1. Create a shell script to automate project directory creation.

```
yashu@LAPTOP-J824BISG:~/MarketingAnalytics$ chmod +x create_project.sh
yashu@LAPTOP-J824BISG:~/MarketingAnalytics$ ./create_project.sh
./create_project.sh: line 1: Project: command not found
yashu@LAPTOP-J824BISG:~/MarketingAnalytics$ cat create_project.sh
Project directories created successfully.
```

2. Find the top 10 largest files in the system.

```
77 ./Reports/Marketing_Report.txt
77 ./Reports/Marketing_Report.txt
73 ./Reports/customer_data.txt
42 ./create_project.sh
```

3. Schedule a backup task using cron.

```
yashu@LAPTOP-J824BISG:~/MarketingAnalytics$ crontab -e
no crontab for yashu - using an empty one

Select an editor.  To change later, run 'select-editor'.
  1. /bin/nano        <---- easiest
  2. /usr/bin/vim.basic
  3. /usr/bin/vim.tiny
  4. /bin/ed

Choose 1-4 [1]: 1
crontab: installing new crontab
yashu@LAPTOP-J824BISG:~/MarketingAnalytics$ contrab -l
```

4. Monitor CPU utilization continuously using top/htop.

```
yashu@LAPTOP-J824BISG:~/MarketingAnalytics$ top
top - 17:26:31 up 56 min,  1 user,  load average: 0.01, 0.22, 0.18
Tasks:  23 total,   1 running,  22 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.1 sy,  0.0 ni, 99.9 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :  11776.2 total,  11155.5 free,    562.4 used,    233.0 buff/cache
MiB Swap:   3072.0 total,   3072.0 free,      0.0 used.  11213.8 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
      1 root      20   0   21744  12384   9440 S   0.0   0.1   0:00.99 systemd
      2 root      20   0    3120   1920   1920 S   0.0   0.0   0:00.01 init-systemd(Ub
      7 root      20   0    3120   1792   1792 S   0.0   0.0   0:00.00 init
     46 root      19  -1   66828  18892  17996 S   0.0   0.2   0:00.50 systemd-journal
     93 root      20   0   25140   6144   4864 S   0.0   0.1   0:01.23 systemd-udevd
    111 systemd+  20   0   21336  12032  10368 S   0.0   0.1   0:00.09 systemd-resolve
    112 systemd+  20   0   91028   7552   6784 S   0.0   0.1   0:00.12 systemd-timesyn
    180 root      20   0    4236   2432   2304 S   0.0   0.0   0:00.01 cron
    181 message+  20   0    9640   4992   4480 S   0.0   0.0   0:00.14 dbus-daemon
    194 root      20   0   17964   8064   7296 S   0.0   0.1   0:00.11 systemd-logind
    202 root      20   0    3160   1920   1792 S   0.0   0.0   0:00.01 agetty
    217 syslog    20   0  222508   5504   4352 S   0.0   0.0   0:00.07 rsyslogd
    220 root      20   0    3116   1792   1664 S   0.0   0.0   0:00.00 agetty
    226 root      20   0  107028  22528  13184 S   0.0   0.2   0:00.10 unattended-upgr
    313 root      20   0    6692   4352   3712 S   0.0   0.0   0:00.00 login
    367 yashu     20   0   20316  11264   9216 S   0.0   0.1   0:00.10 systemd
    368 yashu     20   0   21152   3512   1792 S   0.0   0.0   0:00.00 (sd-pam)
    394 yashu     20   0    6072   5120   3584 S   0.0   0.0   0:00.02 bash
    957 polkitd   20   0  308164   7680   6912 S   0.0   0.1   0:00.06 polkitd
   1496 root      20   0    3124    904    768 S   0.0   0.0   0:00.00 SessionLeader
   1497 root      20   0    3140   1164   1024 S   0.0   0.0   0:00.03 Relay(1498)
   1498 yashu     20   0    6072   4992   3456 S   0.0   0.0   0:00.04 bash
   1627 yashu     20   0    9408   5504   3328 R   0.0   0.0   0:00.01 top
```

5. Generate a disk usage report for all project directories.

```
yashu@LAPTOP-J824BISG:~/MarketingAnalytics$ du -sh ~/MarketingAnalytics/*
4.0K    /home/yashu/MarketingAnalytics/ProcessedData
4.0K    /home/yashu/MarketingAnalytics/RawData
16K     /home/yashu/MarketingAnalytics/Reports
4.0K    /home/yashu/MarketingAnalytics/create_project.sh
```

Tools /Technology used:

- Ubuntu Linux (WSL)

- Bash Shell
- Linux Terminal
- Nano Editor
- Cron
- Core Linux Commands (whoami, pwd, ls, mkdir, touch, cat, grep, find, cp, mv, rm, rmdir, du, top)

Conclusion:

This practical provided hands-on experience with essential Linux commands for system administration. It demonstrated how to create and manage files and directories, search data, perform backup operations, monitor system resources, and automate tasks using shell scripts and cron, building a strong foundation in Linux-based file management and administration.